UE24CS320A – Capstone Project Approval

Project Title	:	A Comparative Framework for Early Sepsis Prediction Using Ensemble and Explainable Deep Learning on ICU Data
Project ID	:	186
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Problem Statement

Sepsis mortality rises with every hour treatment is delayed, so predicting onset 6 to 12 hours early is critical. Machine-learning models predict well and SHAP is the default explanation, yet no one checks whether those explanations can be trusted.

1. Unfair model comparison

Ensemble models (AUROC 0.85–0.91) and LSTM (about 0.85) are reported on different datasets, splits and labels, so no fair comparison exists.

2. Explanations never checked

Features are correlated and labs are 85–95% missing, yet no study tests whether SHAP is stable, faithful and robust.

3. Unstable, with no warning

Theory says no ranking can be faithful, stable and complete at once. Clinicians still get explanations with no reliability indicator.

Objective: (i) compare XGBoost, Random Forest and LSTM under identical preprocessing, patient-level splits and evaluation; (ii) build a reliability layer that detects, diagnoses and mitigates unreliable SHAP explanations, and abstains when reliability stays too low.

Scope and Feasibility Study

In scope

- PhysioNet/CinC 2019 (~40,336 ICU patients)
- XGBoost, Random Forest, LSTM
- TreeSHAP, Kernel SHAP, Integrated Gradients
- Stability, faithfulness, robustness
- Detect, diagnose, mitigate, abstain

Out of scope

- New model architectures
- Chasing AUROC records
- Prospective clinical validation
- Multimodal data (imaging, waveforms)
- Real-time hospital deployment

Feasibility

Implementation: the layer wraps existing models; SHAP, Captum, XGBoost and PyTorch are mature.
Evaluation: failure labels are defined before the detector, so evaluation is not circular.
Systematic: perfect reliability is impossible under collinearity, so we detect rather than eliminate.
Practical: a post-hoc layer, not real-time; latency and cost are acceptable.

Challenges → Mitigation

No benchmark for explanation reliability → define independent failure labels first
Labs 85–95% missing → compare imputed vs dropped explanations
Correlated features → dependence-aware attribution (Latent SHAP)
Circular evaluation → labels from retraining, perturbation and explainer disagreement

Background

0.85–0.91

AUROC of ensemble models on PhysioNet 2019; LSTM reaches about 0.85

PhysioNet/CinC Challenge 2019 studies

85–95%

missingness in lab values, so SHAP depends on which features were measured

PhysioNet/CinC 2019 data

+67% / yr

growth in explainability adoption across 37 sepsis studies; PCT/CRP rarely rank high

BMC Med. Inform. Decis. Mak., 2026

Domain context

Early sepsis prediction uses ensemble models and deep learning, and SHAP is the default explanation. Because clinical variables are correlated, theory (Attribution Impossibility, 2026) shows no feature ranking can be faithful, stable and complete together. A clinician may see respiratory rate as the top feature, then heart rate after retraining. The prediction is stable, the explanation is not, and nothing tells the clinician.

Background Work

Work	What it shows	What is missing
Ensemble models (XGBoost, RF, LightGBM)	AUROC 0.85–0.91 on PhysioNet 2019	Compared with deep learning only across different pipelines
LSTM-GAT; MIMIC-Sepsis benchmark (2025–26)	AUROC 0.853 ± 0.005; Transformer 0.919	No explanation reliability evaluation
XAI for sepsis review, 37 studies (2026)	Adoption +67% per year; key biomarkers rarely rank high	Diagnoses the gap, offers no solution
SHAP failure theory (Huang 2024; Kumar 2020)	SHAP can mislead; many Shapley variants differ	Theoretical; no sepsis or clinical framework
Latent SHAP, ReCalX, UbiQTree (2024–25)	Each fixes or quantifies one failure mode	Not a detector; not applied to sepsis
Our framework	Unified comparison + detect, diagnose, mitigate, abstain	—

Research gap: no study evaluates whether SHAP explanations for sepsis are stable, faithful and robust, or abstains when they are not.

Proposed Approach



Blue boxes = our contribution. Output: a reliable explanation, or an "Explanation unreliable" warning.

Reliability score

$$R(x) = w_s \cdot \text{Stability}(x) + w_f \cdot \text{Faithfulness}(x) + w_r \cdot \text{Robustness}(x)$$

Weights start at 1/3 each and are tuned on the validation set.

Example output

"Prediction: 82% sepsis risk. Explanation reliability: LOW. Top feature ranking is unstable across model retraining. Use clinical judgment."

Stack: Python | XGBoost / scikit-learn | PyTorch | SHAP | Captum | SciPy | Pandas / NumPy | Streamlit (optional)

Reliability Layer: Detect

Metric	How it is computed	Flag when
A. Stability across retraining	Train the same model with 10 seeds; compare SHAP for the same patient using Spearman rank correlation and top-k Jaccard overlap	Jaccard < 0.6
B. Faithfulness	Compare top-K overlap between SHAP and Integrated Gradients for the same patient	Strong disagreement
C. Robustness	Perturb one input within a plausible range (e.g. heart rate ±5 bpm); compare cosine similarity, Euclidean distance and change rate	Cosine < 0.8
D. Temporal stability (LSTM)	Check that important time steps stay consistent at 6 h, 8 h and 10 h before onset	Window shifts

Independent failure labels (defined before the detector is built)

- **Retraining instability**
- **Perturbation sensitivity**
- **Missingness sensitivity**
- **Explainer disagreement**

Diagnose, Mitigate and Abstain

Failure mode	Diagnosis	Mitigation
Feature correlation	Pairwise correlation among top-K features	Dependence-aware attribution (Latent SHAP)
Missingness sensitivity	Compare imputed vs dropped explanations	Informative missingness modelling
Model instability	Retraining variance	Ensemble attribution across models
Perturbation sensitivity	Explanation change under small input changes	ReCalX-style calibration
Temporal instability	Important time window shifts	DeltaSHAP (explains changes between time steps)

Stage 4 – Abstain

If $R(x) < 0.5$, show the risk score with a LOW-reliability warning instead of a feature ranking: “Top feature ranking is unstable across model retraining. Use clinical judgment.”

Applications / Use Cases

1. ICU decision support
A reliability label sits beside every sepsis risk score, so doctors know when not to over-rely on feature attributions.

2. Model auditing
AI governance teams audit deployed sepsis models regularly and catch explanation drift over time.

3. XAI method comparison
Researchers evaluate how reliable different explanation methods are on clinical time-series data.

4. Regulatory support
Quantitative reliability evidence for explainability requirements in AI medical-device submissions.

5. Reusable pattern
Detect, diagnose, mitigate, abstain applies to any clinical prediction task with correlated features and missing data.

Expected Deliverables

Capstone-I – Foundation

- Literature review and novelty matrix
- Dataset preparation and exploratory analysis
- Baselines: XGBoost, RF, LSTM
- Explanation pipeline: SHAP, Integrated Gradients
- Feasibility study and evaluation plan

Milestone: Baselines and explanations running

Capstone-II – Core novelty

- Reliability metrics: stability, faithfulness, robustness
- Independent failure labels defined and generated
- Detector built and evaluated
- Diagnosis and mitigation strategies
- Low-Level Design (LLD)

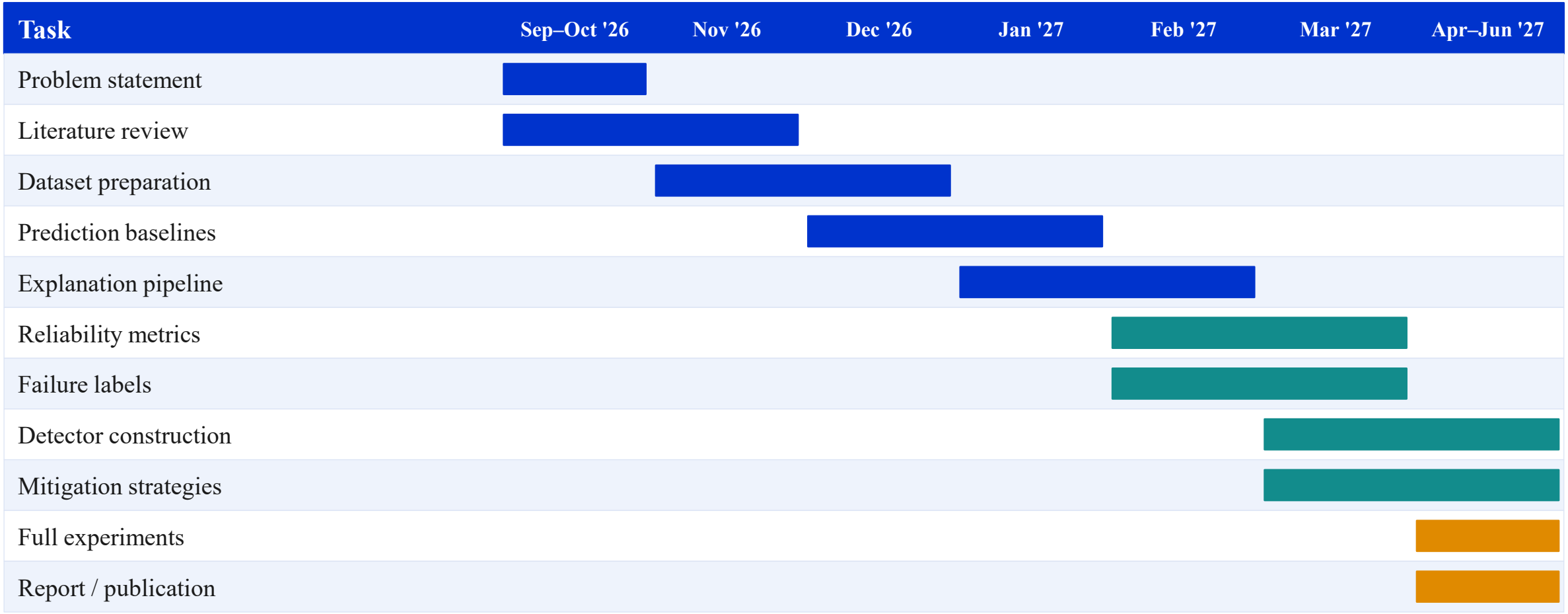
Milestone: Working reliability detector

Capstone-III – Evaluation and paper

- Complete implementation and experiments
- Full evaluation and ablations
- Statistical analysis and visualisation
- Research paper and final documentation
- Reproducibility materials

Milestone: End-to-end framework and paper

Capstone Project Timeline



Blue = Capstone-I (foundation) | Teal = Capstone-II (core novelty) | Orange = Capstone-III (evaluation and paper)

Effort Split by Team Member

Member / Role	Phase-I	Phase-II	Phase-III	Share
Kiran BG **Prediction models and comparison**	Study ensemble and deep-learning baselines; define evaluation protocol	Implement XGBoost, RF, LSTM; run unified comparison	Analyse prediction performance; ablations; documentation	25%
Mohammad Anan **Explanations and reliability metrics**	Study SHAP failure modes; define stability, faithfulness, robustness	Implement SHAP / IG pipelines; compute reliability metrics	Tune metrics; analyse failure patterns; final evaluation	25%
Ibrahim Khaleel **Failure labels and detector**	Define independent failure labels; design detection strategy	Generate failure annotations; build and evaluate detector	Evaluate detection across scenarios	25%
Nikhil KR **Diagnosis, mitigation, integration**	Study failure-mode diagnosis and mitigation	Implement diagnosis, mitigation and abstention	Integrate modules; end-to-end testing	25%

Shared by all four: literature review, own section of the paper, joint evaluation, review presentations and final demo.

Any Other Information: Data and Evaluation

Dataset	What it provides	Used for
PhysioNet/CinC 2019 (primary)	40,336 adult ICU patients, 2 US hospitals; 40 variables, hourly, Sepsis-3 labels; labs 85–95% missing	Training, model comparison, reliability analysis
MIMIC-IV (if feasible)	76,307 ICU patients	External validation
Reliability annotations (ours)	Stability, faithfulness, robustness scores and failure labels per model-explainer pair	Studying SHAP failures; training the detector

Prediction
AUROC, AUPRC, sensitivity, specificity, calibration

Reliability detection
Precision, recall, F1, AUROC for failure detection

Protocol
Patient-level splits, training-only preprocessing, no future-data leakage

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Thank You

Predict sepsis early, check whether the explanation can be trusted, and say so when it cannot.